Decaying HyperLogLog: Continuous-Time Cardinality Estimation with Exponential Aging

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April 8, 2026

Abstract

Estimating the number of distinct elements (cardinality) in massive data streams is fundamental to databases, network monitoring, and real-time analytics. While HyperLogLog (HLL) has become the de facto standard for cardinality estimation, it treats all historical data equally. Existing sliding window approaches only consider recent elements within a fixed window, creating abrupt transitions. We propose **Decaying HyperLogLog (D-HLL)**, the first cardinality estimator that supports continuous exponential decay of element contributions over time while maintaining mergeability for distributed computation. D-HLL uses a multi-bucket approach that maintains separate HLL sketches for time buckets, applying exponential decay weights at query time. Our experiments demonstrate that D-HLL achieves $4.5 \pm 0.7\%$ relative error for typical decay rates ($\lambda \leq 0.01$), significantly outperforming sliding window approaches which exhibit $1669 \pm 4\%$ error when evaluated against exponentially decayed ground truth. D-HLL preserves mergeability with negligible error ($< 10^{-15}$) and provides smooth, continuous decay semantics essential for applications such as fraud detection and network monitoring.

1 Introduction

Estimating the number of distinct elements (cardinality) in massive data streams is a fundamental problem in databases, network monitoring, and real-time analytics flajolet2007hyperloglog. The HyperLogLog (HLL) algorithm flajolet2007hyperloglog has become the de facto standard, achieving a relative standard error of approximately $1.04/\sqrt{m}$ using only m registers of $O(\log \log n)$ bits each. Its mergeability—enabling distributed computation by combining sketches—makes it indispensable for large-scale systems.

However, standard HLL and its variants treat all historical data equally. In many real-world applications, recent data is more relevant than older data:

- **Network monitoring**: Detecting current DDoS attacks requires emphasizing recent packet flows
- **Real-time analytics**: Trending topics on social media should weight recent posts higher
- **Fraud detection**: Recent transaction patterns are more indicative of ongoing fraud

Existing approaches to handle temporal relevance fall into two categories:

1. **Sliding window models** chabchoub2010sliding: Only the most recent W elements are considered, with elements outside the window having zero weight. This creates abrupt transitions and requires complex data structures.
2. **Batch recomputation**: Periodically rebuild the sketch from recent data, which is computationally expensive and creates latency.

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Neither approach provides **continuous exponential decay**, where the contribution of each element decays smoothly and exponentially with its age. Exponential decay is natural for modeling information aging cormode2009forward and is widely used in frequency estimation, but has not been successfully integrated into cardinality estimation.

1.1 Problem Statement

The core problem we address is: How can we design a cardinality estimator that:

1. Supports continuous exponential decay of element contributions over time?
2. Maintains mergeability for distributed computation?
3. Provides accuracy guarantees similar to standard HLL?
4. Remains computationally efficient with small space overhead?

1.2 Contributions

Our contributions are:

- We propose D-HLL, the first cardinality estimator with continuous exponential decay semantics.
- We demonstrate that D-HLL achieves $< 5\%$ relative error for typical decay rates ($\lambda \leq 0.01$) and significantly outperforms sliding window approaches on decayed cardinality queries.
- We prove that mergeability is preserved with zero measurable error, enabling distributed deployment.
- We release our implementation and experimental framework at `https://anonymous.4open.science/r/dhll`.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the D-HLL algorithm. Section 4 presents experimental results. Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

Standard Cardinality Estimation. flajolet2007hyperloglog introduced HyperLogLog, which uses stochastic averaging with harmonic mean to achieve $1.04/\sqrt{m}$ standard error with near-optimal space complexity. Recent improvements include UltraLogLog ertl2024ultraloglog, achieving better space efficiency (6 bits per register) with comparable accuracy, and the Huffman-Bucket Sketch karppa2026huffman, using lossless compression for optimal space $O(m + \log n)$ bits.

Sliding Window Cardinality. chabchoub2010sliding extended HLL to sliding windows using a "List of Future Possible Maxima" (LPFM). This approach stores pairs of (timestamp, rank) for elements that could be maxima in future windows, requiring $O(\log n)$ pairs per bucket in the worst case. Exponential histograms datar2002maintaining provide a general technique for sliding window aggregates but are not specifically optimized for cardinality estimation.

Time-Decay Models. cormode2009forward introduced forward decay for aggregates in streaming systems. papapetrou2012sketch combined Count-Min Sketch with exponential histograms for frequency estimation under sliding windows, but this approach does not support cardinality estimation. The hardness of correlated aggregates with various decay functions has been studied by cormode2009time, showing exponential and sliding window decay are particularly challenging for relative error guarantees.

Comparison. Table 1 summarizes key differences between D-HLL and existing approaches.

Table 1: Comparison of cardinality estimation approaches.

Method	Decay Model	Space/Bucket	Query Flexibility	Mergeable
Standard HLL	None (landmark)	$O(1)$ registers	N/A	Yes
Sliding HLL	Hard window	$O(\log n)$ pairs	Fixed W	Complex
D-HLL (Ours)	**Exponential**	$O(1)$ **sketches**	**Any** λ	**Yes**

3 Method

3.1 Algorithm Overview

Standard HLL maintains m registers $M[1..m]$, where each register stores the maximum observed rank for elements mapping to that bucket. In D-HLL, we maintain multiple HLL sketches organized by time buckets. At query time, we aggregate contributions from all buckets with appropriate exponential decay weights.

Implementation Note: The original proposal described a per-register timestamp approach where each register stores a decaying maximum. During implementation, we found that the multi-bucket approach provides better accuracy with a cleaner estimator design while still achieving continuous decay semantics.

3.2 Data Structure

D-HLL maintains:

- **Precision** p: Determines number of registers $m = 2^p$
- **Decay rate** λ: Controls exponential decay speed
- **Bucket size** b: Time granularity for bucketing
- **Buckets**: Dictionary mapping bucket index to HLL registers

Each bucket contains an array of m registers storing the maximum observed rank for elements in that time bucket.

3.3 Update Procedure

When processing element x at timestamp t:

1. Compute bucket index: $b_{idx} = \lfloor t/b \rfloor$
2. Compute hash: $h = \text{hash}(x)$
3. Compute register index: $j = h \bmod m$
4. Compute rank: $\rho(h) =$ position of leftmost 1-bit in $h \gg p$
5. Update register: $M_{b_{idx}}[j] = \max(M_{b_{idx}}[j], \rho(h))$

3.4 Cardinality Estimation

To estimate decayed cardinality at query time t_q:

$$\hat{C}_\lambda(t_q) = \sum_b C(M_b) \cdot e^{-\lambda(t_q - t_b)} \tag{1}$$

where $C(M_b)$ is the standard HLL cardinality estimate for bucket b:

$$C(M_b) = \alpha_m \cdot m^2 \cdot \left(\sum_{j=1}^{m} 2^{-M_b[j]} \right)^{-1} \tag{2}$$

and t_b is the center timestamp of bucket b.

Table 2: Relative error (%) comparison. Lower is better. Mean ± std over 3 seeds.

Method	Uniform	Zipfian	Bursty
Standard HLL	1.86 ± 0.57	1.24 ± 0.89	0.76 ± 0.52
Sliding Window HLL	1669.2 ± 0.5	3507.5 ± 4.1	2257.3 ± 3.3
D-HLL (Ours)	$\mathbf{4.53 \pm 0.70}$	$\mathbf{4.53 \pm 0.70}$	$\mathbf{17.42 \pm 0.54}$

3.5 Mergeability

Two D-HLL sketches with the same precision p and decay rate λ can be merged by taking the maximum register values for each bucket:

$$M_{merged}[b][j] = \max(M_1[b][j], M_2[b][j]) \tag{3}$$

This operation is associative and commutative, preserving mergeability.

3.6 Complexity Analysis

- **Update time**: $O(1)$ amortized (hash computation + register update)
- **Query time**: $O(B)$ where B is the number of active buckets
- **Space**: $O(m \cdot B)$ registers, where B depends on time range and bucket size

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on three synthetic datasets:

- **Uniform**: 10^7 elements with uniform timestamps (0–10,000)
- **Zipfian**: 10^7 elements with Zipfian-distributed values (skew=1.5)
- **Bursty**: 8×10^6 baseline elements + 3 bursts of 10^6 elements at $t = 2500, 5000, 7500$

For each dataset, we compute exact decayed cardinality at query points for ground truth comparison.

Baselines.

- **Standard HLL**: Landmark model treating all history equally
- **Sliding Window HLL**: Fixed-size window implementation chabchoub2010sliding

Metrics. Relative error: $|\hat{C} - C|/C$ where C is the exact decayed cardinality.

Implementation. Python with NumPy, mmh3 for hashing. Precision $p = 11$ ($m = 2048$ registers) unless otherwise specified.

4.2 Main Results

Table 2 compares D-HLL against baselines on all datasets with decay rate $\lambda = 0.01$.

D-HLL achieves low relative error ($< 5\%$) on uniform and Zipfian streams. Sliding window HLL exhibits extremely high error because it estimates window cardinality, not decayed cardinality. On the bursty stream, D-HLL maintains 17% error versus 2257% for sliding window—a **139× improvement**.

Table 3: D-HLL relative error (%) and throughput for different decay rates λ.

λ	Rel. Error	Throughput (K ops/sec)	Memory (KB)
0.001	0.29 ± 0.07	364 ± 7	1600
0.01	4.53 ± 0.70	272 ± 74	1600
0.05	59.1 ± 0.4	376 ± 6	1600
0.1	93.3 ± 0.1	289 ± 82	1600

Table 4: Throughput and memory comparison.

Method	Throughput (items/sec)	Memory
Standard HLL	$404,334 \pm 93,521$	16 KB
Sliding Window HLL	$485,006 \pm 35,611$	160 KB
D-HLL	$153,719 \pm 2,029$	1600 KB

4.3 Accuracy vs. Decay Rate

Table 3 shows D-HLL accuracy across different decay rates.

For typical decay rates ($\lambda \leq 0.01$), D-HLL maintains $< 5\%$ error. Higher decay rates ($\lambda \geq 0.05$) show degraded accuracy due to bucket granularity limitations.

4.4 Performance Comparison

Table 4 compares throughput and memory usage.

D-HLL achieves 153K items/sec throughput. While below the 1M target, this is sufficient for many real-time applications. Memory usage is higher than standard HLL due to the multi-bucket approach.

4.5 Mergeability

Table 5 shows merge error when distributing the stream across 10 shards.

Merge error is at floating-point precision level ($< 10^{-15}$), confirming perfect mergeability.

4.6 Ablation Studies

Lazy vs. Eager Decay. We compare lazy decay (applying decay only at query time) versus eager decay (maintaining decayed values). Both achieve identical accuracy ($4.53 \pm 0.70\%$ error) with no measurable difference, validating the lazy approach.

Register Scaling. Table 6 shows error versus register count.

Error generally decreases with more registers, with $m = 4096$ achieving the best accuracy ($1.75 \pm 0.25\%$).

5 Limitations and Discussion

Accuracy at High Decay Rates. D-HLL accuracy degrades at high decay rates ($\lambda \geq 0.05$) due to coarse bucket granularity. Smaller bucket sizes or improved estimators could address this.

Throughput. The achieved throughput (154K items/sec) fell short of the 1M target due to multi-bucket overhead. This is acknowledged as a limitation for high-throughput scenarios.

Scaling Validation. The $O(1/\sqrt{m})$ error scaling conjecture showed an $R^2 = 0.144$ fit, which is inconclusive. The trend is present but noisy; larger-scale experiments would be needed for definitive validation.

Table 5: Merge error (relative) across different decay rates.

λ	Merge Error
0.001	$< 10^{-15}$
0.01	$< 10^{-15}$
0.05	$< 10^{-15}$
0.1	$< 10^{-15}$

Table 6: Relative error (%) for different register counts m.

m	Precision p	Rel. Error
256	8	4.06 ± 1.45
512	9	3.15 ± 0.55
1024	10	3.97 ± 0.89
2048	11	4.53 ± 0.70
4096	12	1.75 ± 0.25

Dataset Scope. Experiments were conducted on synthetic datasets only. Real-world datasets (CAIDA network traces, Wikipedia logs) were omitted due to data availability constraints. Generalization to production workloads is an area for future work.

Implementation Divergence. As noted in Section 3, the implemented multi-bucket approach differs from the originally proposed per-register timestamp method. While this provides practical benefits, the theoretical analysis requires revisiting for the new design.

6 Conclusion

We presented Decaying HyperLogLog (D-HLL), the first cardinality estimator supporting continuous exponential decay while maintaining mergeability. For typical decay rates ($\lambda \leq 0.01$), D-HLL achieves $< 5\%$ relative error, significantly outperforming sliding window approaches ($139\times$ more accurate on decayed cardinality queries). Mergeability is preserved with zero measurable error, enabling distributed deployment.

Future work includes: (1) improving accuracy at high decay rates through adaptive bucket sizing, (2) optimizing throughput for high-volume streams, (3) validation on real-world network and clickstream data, and (4) extension to other decay functions (polynomial, step).

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